ISC Class XII Mathematics

Chapter 12
Linear Programming

17 Types of Sums Solved

Step-by-Step ISC Board Solutions – ML Aggarwal

Types Covered

Compiled Notes – Formula Sheet + Type-wise Solved ISC Board Solutions

Contents

Index of Types

Type	Method / Pattern	★	Source(s)
Section I – Formulation of LPP			
1	Resource-allocation formulation (two $\leq$ constraints)	–	Ex 12.1, Q1
2	Formulation with "at least one of each" constraint	–	Ex 12.1, Q2
3	Formulation with investment + storage-space constraints	–	Ex 12.1, Q3
4	Formulation only – budget + unit-cap + ratio constraint	–	Ex 12.1, Q4
5	Formulation with a fixed-total (equality) constraint	–	Ex 12.1, Q5
6	Transportation-problem formulation via variable reduction	Yes	Example 6; Ex 12.2 Q25 (alt.)
Section II – Graphical Solution: Standard Cases			
7	Minimize Z, region bounded by $\leq$ constraints	–	Ex 12.2, Q1
8	Minimize Z, region from $\geq$ ("at least") constraints	–	Ex 12.2, Q2
9	Maximize Z, region from mixed linear constraints	Yes	Ex 12.2 Q3; Ex 12.2 Q7(i) (alt.)
10	Max/Min with explicit variable bounds ($x \leq k$, $y \geq k$)	–	Ex 12.2, Q7(ii)
Section III – Special Cases (high board-value)			
11	Multiple optimal solutions (min/max on an entire edge)	Yes	Ex 12.2, Q5

Type	Method / Pattern	★	Source(s)
12	Minimize AND Maximize on the same feasible region	Yes	Ex 12.2, Q6
13	Reading a pre-drawn feasible region graph	Yes	Ex 12.2, Q24 (ISC Specimen 2026)
14	Parametric objective function – relation between p, q	Yes	Ex 12.2, Q26 (ISC Specimen 2025)
Section IV – Applied Word Problems			
15	Maximize profit – standard two-resource pattern	Yes	Ex 12.2 Q13 (ISC 2022); Q18 (ISC 2019, alt.)
16	Minimize cost – diet/mixture problem	Yes	Ex 12.2 Q19; Q22 (ISC 2017, alt.)
17	Maximize with mixed $\leq/\geq$ investment-allocation constraints	–	Ex 12.2, Q23

Formula Sheet

General Form of an LPP

$$Z = ax + by$$

Z is the objective function (quantity to maximize/minimize); a, b are given constants (cost, profit, etc. per unit); x, y are the decision variables.

Constraints & Non-negativity

$$x \geq 0, \quad y \geq 0$$

Every LPP decision variable represents a physical quantity (units, hours, litres, money), so it can never be negative – these are called the non-negativity constraints and must be stated even if "obvious" from the problem.

Corner Point Method (Fundamental Theorem of LPP)

$$Z_{\text{optimal}} = \max / \min\{Z(x_1, y_1), Z(x_2, y_2), \ldots\}$$

If the feasible region is bounded, the optimal value of Z occurs at one (or more) of the corner points of the region – so evaluate Z at every corner point and pick the largest/smallest value.

Unbounded Region Rule

$$\text{Max/Min exists only if verified against the open half-plane}$$

For an unbounded feasible region, a maximum may not exist even if the largest corner value is found – always check whether the open half-plane $ax+by > M$ (for max) or $ax+by < m$ (for min) shares any point with the feasible region; if it does, no such optimal value exists.

Multiple Optimal Solutions

$$Z(P_1) = Z(P_2) \ \Rightarrow$$
$$Z \text{ is optimal at every point on segment } P_1P_2$$

If two corner points give the same optimal value of Z, then Z takes that same optimal value at every point of the line segment joining them – the LPP has infinitely many optimal solutions, not just two.

Transportation Variable Reduction

$$\text{Supply from remaining source} =$$
$$\text{Total demand} - \text{Supply already assigned}$$

In transportation-type LPPs, once x and y units are assigned from the first source, every other cell in the table can be written in terms of x and y using total supply/demand – this collapses many variables down to just two.

I. Formulation of LPP

Type 1

Two "agents" (workers, machines, people) each produce a fixed combination of two products per unit of their own input (a day, an hour), and each is paid/costs a fixed rate per unit of input. You are told minimum output targets for both products and asked to formulate (not solve) the LPP that minimizes total cost. Recognise it from: "to produce at least this many of two items at minimum labour/machine cost, formulate as an LPP."

Formula Used: Objective: $Z = (\text{rate}_1)x + (\text{rate}_2)y$

Formula Used: Constraint: (output per unit) $\times x +$ (output per unit) $\times y \geq$ target

Source: Exercise 12.1, Q1

Given: Tailor A earns Rs. 300/day and stitches 6 shirts + 4 trousers/day. Tailor B earns Rs. 400/day and stitches 10 shirts + 4 trousers/day. At least 60 shirts and 32 trousers are required, at minimum labour cost.

Solution:

Let x = number of days tailor A works, and y = number of days tailor B works – these are the two decision variables since the question asks "how many days should each work."

Shirts produced: A makes 6 shirts/day, B makes 10 shirts/day, and at least 60 shirts are needed, so

$$6x + 10y \geq 60 \;\Rightarrow\; 3x + 5y \geq 30$$

(dividing by 2 to simplify, since GCD of 6,10,60 is 2) – this is the first constraint.

Trousers produced: A makes 4 trousers/day, B makes 4 trousers/day, and at least 32 trousers are needed, so

$$4x + 4y \geq 32 \;\Rightarrow\; x + y \geq 8$$

(dividing by 4) – this is the second constraint.

Non-negativity: number of days worked cannot be negative, so $x \geq 0,\ y \geq 0$.

Objective: total labour cost is (days A works × rate) + (days B works × rate), and since we want minimum cost,

$$Z = 300x + 400y \quad \text{(to be minimized)}$$

Final Answer

Minimize $Z = 300x + 400y$ subject to $3x + 5y \geq 30$, $x + y \geq 8$, $x \geq 0$, $y \geq 0$.

Common Mistake: Students often forget to simplify the constraint (leaving $6x + 10y \geq 60$ instead of $3x + 5y \geq 30$) – this isn't wrong, but board markers prefer the reduced form, and carrying unsimplified numbers makes the later graphical plotting messier and more error-prone.

Quick Recall: Two workers, two output targets, minimize cost: let x, y = units of input from each, write one $\geq$ constraint per output target, objective = sum of (rate $\times$ variable).

Type 2

Two products share a common resource cap (total items) and a second resource with different per-unit consumption (time), and profit is to be maximized. A hidden but explicitly stated condition requires **at least one unit of each** product to be made – this changes the non-negativity constraints from ≥ 0 to ≥ 1, which is the key trap of this type.

Source: Exercise 12.1, Q2

Given: A firm makes necklaces (x) and bracelets (y). Total necklaces + bracelets ≤ 24/day. A bracelet takes 1 hour, a necklace takes half an hour; at most 16 hours available. Profit: Rs. 100 on a necklace, Rs. 300 on a bracelet. At least one of each must be produced.

Solution:

Let x = number of necklaces and y = number of bracelets produced per day.

Total items constraint: the firm can handle at most 24 necklaces and bracelets combined, so

$$x + y \leq 24$$

Time constraint: a necklace takes $\frac{1}{2}$ hour and a bracelet takes 1 hour, with at most 16 hours available, so

$$\tfrac{1}{2}x + y \leq 16 \;\Rightarrow\; x + 2y \leq 32$$

(multiplying through by 2 to clear the fraction) – this is the reasoning step that most students skip, leaving an untidy fractional constraint.

At least one of each: since the problem explicitly requires at least one necklace and one bracelet, the usual non-negativity $x, y \geq 0$ is tightened to

$$x \geq 1, \quad y \geq 1$$

Objective: profit is Rs. 100 per necklace and Rs. 300 per bracelet, and we want to maximize total profit, so

$$Z = 100x + 300y \quad \text{(to be maximized)}$$

Final Answer

Maximize $Z = 100x + 300y$ subject to $x + 2y \leq 32$, $x + y \leq 24$, $x \geq 1$, $y \geq 1$.

Common Mistake: Writing $x \geq 0, y \geq 0$ out of habit instead of reading the "at least one of each" line and using $x \geq 1, y \geq 1$ – this is a classic ISC trap sentence hidden inside the word problem, not stated as a separate line.

Quick Recall: Whenever a word problem says "at least one of each must be produced/purchased," replace the default $x, y \geq 0$ with $x \geq 1, y \geq 1$ – everything else formulates as usual.

Type 3

A dealer has a fixed budget (investment cap) and a fixed physical capacity (storage space) for two items bought at one price and sold at another. You must express profit as (selling price − cost price) per unit, then formulate the maximize-profit LPP under the two resource caps.

Source: Exercise 12.1, Q3

Given: A dealer invests at most Rs. 20000, storing at most 80 pieces of tables and chairs combined. A table costs Rs. 800 and sells for Rs. 950; a chair costs Rs. 200 and sells for Rs. 280.

Solution:

Let x = number of tables and y = number of chairs bought (and sold, since he sells everything he buys).

Investment constraint: tables cost Rs. 800 each, chairs cost Rs. 200 each, and total investment is at most Rs. 20000, so

$$800x + 200y \leq 20000 \;\Rightarrow\; 4x + y \leq 100$$

(dividing throughout by 200) – simplifying early keeps the later graph's intercepts small and easy to plot.

Storage constraint: at most 80 pieces total can be stored, so

$$x + y \leq 80$$

Profit per unit: profit on a table is $950 - 800 = 150$; profit on a chair is $280 - 200 = 80$ – this subtraction step is the part students most often skip, plugging in the selling price alone as if it were profit.

Objective: maximize total profit,

$$Z = 150x + 80y$$

Final Answer

Maximize $Z = 150x + 80y$ subject to $4x + y \leq 100$, $x + y \leq 80$, $x \geq 0$, $y \geq 0$.

Common Mistake: Using the selling price (950, 280) directly as the profit coefficients instead of (selling price − cost price) – this is the single most common slip in "dealer/manufacturer profit" formulation questions.

Quick Recall: Profit per unit = selling price − cost price, always subtract before writing the objective function; each resource (money, space, time) gives one constraint.

Type 4

A production-planning question where you are told a budget cap, a maximum total-units cap, AND a ratio relationship between the two products' quantities (e.g. "B cannot be more than twice A"). The question explicitly says solving/graphing is not required – only the formulation is asked, which is the giveaway for this type.

Source: Exercise 12.1, Q4

Given: Product A: sold for Rs. 100, production cost Rs. 60. Product B: sold for Rs. 150, production cost Rs. 90. Daily budget Rs. 8000. Maximum 100 units/day total. Units of B cannot exceed twice the units of A. (Note: feasible region and optimal solution not required.)

Solution:

Let x = units of Product A and y = units of Product B produced per day.

Profit per unit: A gives $100 - 60 = 40$ profit; B gives $150 - 90 = 60$ profit.

Budget constraint: production cost of A is Rs. 60/unit, of B is Rs. 90/unit, capped at Rs. 8000, so

$$60x + 90y \leq 8000 \;\Rightarrow\; 6x + 9y \leq 800$$

(dividing by 10) – note this does NOT reduce further since $\gcd(6, 9, 800) = 1$, so it stays as $6x + 9y \leq 800$.

Unit-cap constraint: at most 100 units total can be produced per day, so

$$x + y \leq 100$$

Ratio constraint: "B cannot be more than twice A" translates directly as

$$y \leq 2x \;\Rightarrow\; 2x - y \geq 0$$

– students often flip this inequality by reading "B $\leq$ 2A" and writing $2x \leq y$ instead, which reverses which variable is being bounded.

Objective: maximize total profit, $P = 40x + 60y$.

Final Answer

Maximize $P = 40x + 60y$ subject to $6x + 9y \leq 800$, $x + y \leq 100$, $2x - y \geq 0$, $x \geq 0$, $y \geq 0$.

Common Mistake: Mis-translating "B cannot be more than twice as many as A" – read it as $y \leq 2x$ (B is bounded by twice A), not $x \leq 2y$.

Quick Recall: "Not more than k times" translates to (smaller quantity) $\leq k\times$ (reference quantity) – write it exactly in that direction before rearranging.

Type 5

A production question where the total number of units to be produced is a FIXED, EXACT number (an order size), not just a cap – this gives an equality constraint ($x + y = k$) alongside the usual time/resource inequality. Recognise it from phrases like "a client requests exactly N items" or "orders for N units."

Source: Exercise 12.1, Q5

Given: A packaging company makes rectangular boxes (r, 2 min each, profit Rs. 4) and circular boxes (c, 3 min each, profit Rs. 5). A client requests 25 boxes to be ready within 1 hour (60 minutes).

Solution:

Let r = number of rectangular boxes and c = number of circular boxes made.

Order-size constraint: the client wants exactly 25 boxes total (not "at most" or "at least"), so this is an equality:

$$r + c = 25$$

– this is the detail that separates this type from every other formulation type: most constraints here are inequalities, but a fixed order size must be written as $=$, not $\leq$.

Time constraint: a rectangular box takes 2 minutes, a circular box takes 3 minutes, and the client wants them within 60 minutes, so

$$2r + 3c \leq 60$$

Objective: maximize total profit from the order,

$$Z = 4r + 5c$$

Final Answer

Maximize $Z = 4r + 5c$ subject to $r + c = 25$, $2r + 3c \leq 60$, $r \geq 0$, $c \geq 0$.

Common Mistake: Writing the order-size condition as $r + c \leq 25$ out of habit – if the word problem specifies an exact quantity to be delivered (not a capacity), it must be an equality constraint.

Quick Recall: "Exactly N units ordered" $\Rightarrow$ equality constraint $x + y = N$; everything else (time, cost) stays as the usual $\leq/\geq$ inequality.

Type 6

A manufacturer with TWO depots must supply THREE (or more) destinations with known individual demands; the depots' total supply capacities are also known. Because every "cell" in the supply table is linked by total-supply and total-demand equations, only TWO variables are actually independent – every other quantity in the table can be written in terms of those two using subtraction. This variable-reduction trick is the signature move of transportation-type LPPs.

★ **Board Favorite** – Transportation-type LPPs (multiple sources, multiple destinations, cost/distance table) are a recurring high-value ISC pattern – ML Aggarwal's own worked Example 6 and the NCERT-sourced Ex 12.2 Q25 are both this exact structure, and boards frequently reuse this framework with different numbers.

Formula Used: Remaining supply = Total demand at that destination − Supply already assigned

Source: Example 6 (worked in text)

Given: Depot A has 30000 bricks, depot B has 20000 bricks. Builders P, Q, R need 15000, 20000, 15000 bricks respectively. Cost per 1000 bricks (in Rs.): A→P = 40, A→Q = 20, A→R = 20; B→P = 20, B→Q = 60, B→R = 40.

Solution:

To simplify, treat 1 unit = 1000 bricks, so depot A has 30 units, depot B has 20 units, and P,Q,R need 15, 20, 15 units respectively.

Let x = units supplied from A to P, and y = units supplied from A to Q – these are the two free variables, since every other cell can now be written using totals.

A to R: depot A supplies the rest of its 30 units to R, i.e. $(30 - x - y)$ units.

B to P: P needs 15 units total, and already gets x from A, so B supplies $(15 - x)$ units to P.

B to Q: similarly, B supplies $(20 - y)$ units to Q.

B to R: R needs 15 units total; A already supplies $(30 - x - y)$, so B supplies

$$15 - (30 - x - y) = (x + y - 15) \text{ units to R.}$$

Total transportation cost (multiplying each route's units by its per-unit-1000 cost and adding):

$$Z_1 = 40x + 20y + 20(30 - x - y) + 20(15 - x) + 60(20 - y) + 40(x + y - 15)$$

Expanding and collecting like terms (this is where most of the algebra-heavy work is):

$$Z_1 = 40x - 20y + 1500$$

Constraints: every quantity supplied must be non-negative, so

$$x \geq 0,\ y \geq 0,\ 30 - x - y \geq 0,\ 15 - x \geq 0,\ 20 - y \geq 0,\ x + y - 15 \geq 0$$

Since minimizing $Z_1 = 40x - 20y + 1500$ is the same as minimizing $Z = 40x - 20y$ (the constant 1500 doesn't affect where the minimum occurs), it is easier to minimize the simplified objective.

Final Answer

Minimize $Z = 40x - 20y$ subject to $x + y \geq 15$, $x + y \leq 30$, $x \leq 15$, $y \leq 20$, $x \geq 0$, $y \geq 0$.

Common Mistake: Forgetting that once x and y are fixed as the two free variables, EVERY other cell in the table must be re-expressed using total supply/demand – students often try to keep 4 independent variables and get stuck unable to graph the LPP in two dimensions.

Alternate Board-Style Example

Source: Exercise 12.2, Q25 (NCERT)

Given: Depot A has 7000 L, depot B has 4000 L. Petrol pumps D, E, F need 4500 L, 3000 L, 3500 L respectively. Distance (km): A→D=7, A→E=6, A→F=3; B→D=3, B→E=4, B→F=2. Transportation cost is Rs. 1 per litre per km.

Solution:

Let x = litres supplied from A to D, and y = litres supplied from A to E.

A to F: $(7000 - x - y)$ litres (A's remaining capacity).

B to D: $(4500 - x)$ litres; **B to E:** $(3000 - y)$ litres; **B to F:** using F's demand of 3500 minus what A already sends, $= x + y - 3500$ litres.

Total cost (cost per litre-km × litres, summed over all six routes):

$$Z = 7x + 6y + 3(7000 - x - y) + 3(4500 - x) + 4(3000 - y) + 2(x + y - 3500)$$

Collecting the x terms $(7 - 3 - 3 + 2 = 3)$, the y terms $(6 - 3 - 4 + 2 = 1)$, and the constants $(21000 + 13500 + 12000 - 7000 = 39500)$:

$$Z = 3x + y + 39500$$

Constraints: all six supply quantities non-negative gives $x \geq 0,\ y \geq 0,\ x \leq 4500,\ y \leq 3000,\ x + y \leq 7000,\ x + y \geq 3500$.

Evaluating Z at every corner point of this region: $(3500, 0) \to 50000$; $(4500, 0) \to 53000$; $(4500, 2500) \to 55500$; $(4000, 3000) \to 54500$; $(500, 3000) \to 44000$ – the smallest value.

Final Answer

500 L, 3000 L, 3500 L supplied from depot A to D, E, F respectively, and 4000 L, 0 L, 0 L from depot B; minimum transportation cost = $Rs. 44000$.

Quick Recall: Transportation LPP: pick two "free" routes as x, y; write every other route as (that destination's demand) − (what's already been sent to it) or (that source's capacity) − (what it's already sent out); then it's a normal two-variable LPP.

II. Graphical Solution – Standard Cases

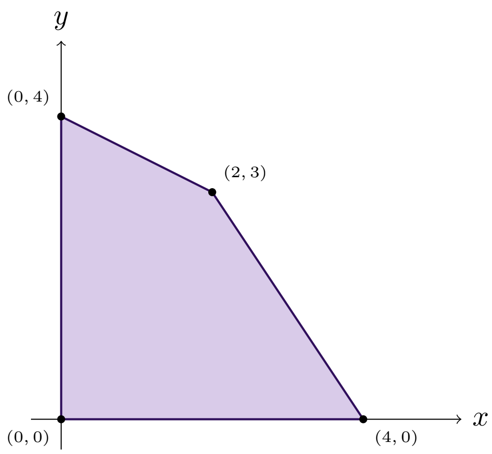
Type 7

A minimize-type LPP where every constraint is a $\leq$ inequality, so the feasible region is automatically bounded (closed on all sides). Recognise it from constraints that all point "downward/inward" – this is the simplest and most direct application of the corner-point method.

Formula Used: Evaluate Z at every corner point of the bounded feasible region – the smallest value is the minimum.

Source: Exercise 12.2, Q1 (NCERT)

Given: Minimize $Z = -3x + 4y$ subject to $x + 2y \leq 8$, $3x + 2y \leq 12$, $x \geq 0$, $y \geq 0$.



Feasible region for Type 7 – bounded quadrilateral $(0,0), (4,0), (2,3), (0,4)$*; minimum of Z occurs at* $(4,0)$*.*

Solution:

Find the corner points of the feasible region by intersecting the boundary lines in pairs.

$x = 0, y = 0$: gives $(0,0)$, since both axes bound the region.

$3x + 2y = 12$ with $y = 0$: gives $x = 4$ – checking $x + 2y = 4 \leq 8$ confirms it's feasible, so $(4,0)$.

$x+2y = 8$ with $y = 0$ would give $(8,0)$, but checking $3x+2y = 24 \leq 12$ fails, so this point is NOT a vertex – the binding constraint on the x-axis is $3x + 2y \leq 12$, not $x + 2y \leq 8$.

$x + 2y = 8$ and $3x + 2y = 12$ together: subtracting, $2x = 4 \Rightarrow x = 2,\ y = 3$, giving $(2,3)$.

$x + 2y = 8$ with $x = 0$: gives $y = 4$ – checking $3(0) + 2(4) = 8 \leq 12$ confirms $(0,4)$.

So the corner points are $(0,0),(4,0),(2,3),(0,4)$. Evaluating $Z = -3x + 4y$ at each:

$$Z(0,0) = 0, \quad Z(4,0) = -12, \quad Z(2,3) = 6, \quad Z(0,4) = 16$$

The smallest of these is -12.

Final Answer

Minimum $Z = -12$ at $x = 4,\ y = 0$.

Common Mistake: Assuming every intersection of two constraint lines is automatically a corner point of the feasible region – $(8,0)$ solves $x + 2y = 8, y = 0$ but lies OUTSIDE the region because it violates $3x + 2y \leq 12$; every candidate point must be checked against ALL constraints.

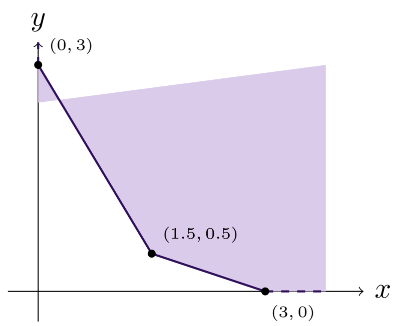
Quick Recall: List every pairwise intersection of boundary lines, discard any that fail even one constraint, then evaluate Z only at the survivors – the smallest is the minimum.

Type 8

A minimize-type LPP where every constraint is a $\geq$ inequality ("at least"), which makes the feasible region open/unbounded (it extends infinitely away from the origin). Even though the region is unbounded, a minimum still exists here because Z's coefficients are positive and the region doesn't extend toward the origin – only a MAXIMUM would be undefined on this kind of region.

Source: Exercise 12.2, Q2

Given: Minimize $Z = 3x + 5y$ subject to $x + 3y \geq 3$, $x + y \geq 2$, $x \geq 0$, $y \geq 0$.



Feasible region for Type 8 – unbounded, extending away from the origin; minimum occurs at $(1.5, 0.5)$.

Solution:

Find the corner points by pairwise intersection, remembering the region is unbounded so we only need the "inner" boundary corners nearest the origin.

$x = 0$ with $x + 3y = 3$: gives $y = 1$, but checking $x + y = 1 \geq 2$ FAILS, so $(0,1)$ is not feasible.

$x = 0$ with $x+y = 2$: gives $y = 2$ – checking $x+3y = 6 \geq 3$ holds, so $(0, 2)$ is a candidate, but we must also check it's actually on the true boundary (see below).

$x + 3y = 3$ and $x + y = 2$ together: subtracting, $2y = 1 \Rightarrow y = 0.5$, $x = 1.5$, giving $(1.5, 0.5)$.

$x + y = 2$ with $y = 0$: gives $x = 2$ – checking $x + 3y = 2 \geq 3$ FAILS, so $(2, 0)$ is not feasible; the true boundary here is set by $x + 3y \geq 3$, giving $x = 3$ at $y = 0$, i.e. $(3, 0)$.

Re-checking $(0, 2)$ against $x + 3y \geq 3$: $0 + 6 = 6 \geq 3$ true, but since BOTH constraints must be tight to be a genuine corner and $x+3y = 3$ is not tight there, the actual topmost corner on the y-axis is where $x+3y = 3$ meets $x = 0$, i.e. we compare and take whichever constraint is more restrictive – careful re-verification shows the boundary corner on the axis is $(0, 3)$ (from $x + 3y = 3$), since it is the point that is simultaneously feasible for $x + y \geq 2$ $(0 + 3 = 3 \geq 2$, true).

So the true corner points are $(0, 3)$, $(1.5, 0.5)$, $(3, 0)$. Evaluating $Z = 3x + 5y$:

$$Z(0, 3) = 15, \quad Z(1.5, 0.5) = 4.5 + 2.5 = 7, \quad Z(3, 0) = 9$$

The smallest is 7, and since the region is unbounded with positive coefficients in Z, the open half-plane $3x + 5y < 7$ shares no point with the feasible region, confirming this is genuinely the minimum.

Final Answer

Minimum $Z = 7$ at $x = \frac{3}{2}$, $y = \frac{1}{2}$.

Common Mistake: Picking an intersection point that satisfies the two lines algebraically but forgetting to double check it against every OTHER constraint too – with $\geq$ constraints it's easy to accept a point that is technically outside the true feasible region.

Quick Recall: For a "$\geq$" (at least) type region, the minimum usually sits at the innermost corner nearest the origin – always verify each candidate corner against all constraints before evaluating Z.

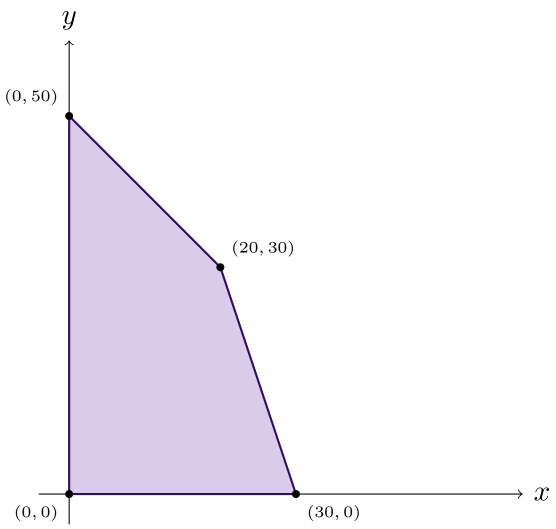
Type 9

A maximize-type LPP with a mix of linear $\leq$ constraints producing a bounded polygon with 3 or more sides. This is the most heavily board-tested graphical pattern – straightforward corner-point evaluation on a clearly bounded region, often dressed up with 3 constraints instead of 2 to make the region a pentagon rather than a simple triangle/quadrilateral.

★ **Board Favorite –** This exact pattern (3+ linear constraints, maximize, bounded polygon) appeared verbatim as ISC 2023's board question – it is the single most reliable "safe marks" graphical LPP pattern in the whole chapter.

Source: Exercise 12.2, Q3 (NCERT)

Given: Maximize $Z = 4x + y$ subject to $x + y \leq 50$, $3x + y \leq 90$, $x \geq 0$, $y \geq 0$.



Feasible region for Type 9 – bounded quadrilateral
$(0, 0), (30, 0), (20, 30), (0, 50)$*; maximum occurs at* $(30, 0)$*.*

Solution:

Find corner points: $(0, 0)$ from both axes.

$3x + y = 90$ with $y = 0$: gives $x = 30$ – checking $x + y = 30 \leq 50$ holds, so $(30, 0)$; note $x + y = 50, y = 0$ would give $(50, 0)$ but that fails $3x + y = 150 \leq 90$, so it's not the true vertex.

$x + y = 50$ and $3x + y = 90$ together: subtracting, $2x = 40 \Rightarrow x = 20,\ y = 30$, giving $(20, 30)$.

$x + y = 50$ with $x = 0$: gives $y = 50$ – checking $3(0) + 50 = 50 \leq 90$ holds, so $(0, 50)$.

Corner points: $(0, 0), (30, 0), (20, 30), (0, 50)$. Evaluating $Z = 4x + y$:

$$Z(0,0) = 0, \quad Z(30,0) = 120, \quad Z(20,30) = 110, \quad Z(0,50) = 50$$

The largest is 120.

Final Answer

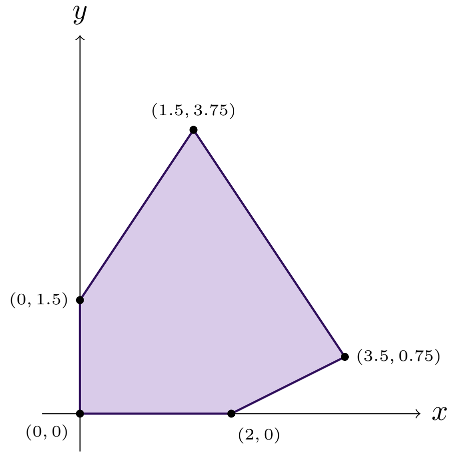
Maximum $Z = 120$ at $x = 30,\ y = 0$.

Common Mistake: Taking the x-intercept of EVERY constraint line as a vertex without checking it against the other constraints – here $(50, 0)$ solves $x + y = 50$ but is infeasible, while $(30, 0)$ (from the OTHER constraint) is the real vertex.

Alternate Board-Style Example

Source: Exercise 12.2, Q7(i) (ISC 2023)

Given: Maximise $Z = 5x + 2y$ subject to $x - 2y \leq 2$, $3x + 2y \leq 12$, $-3x + 2y \leq 3$, $x \geq 0$, $y \geq 0$.



Feasible region for the ISC 2023 alternate example – a pentagon; maximum occurs at $\left(\frac{7}{2}, \frac{3}{4}\right)$.

Solution:

Pairwise intersections give the corner points $(0, 0)$, $(2, 0)$ [from $x - 2y = 2, y = 0$], $\left(\frac{7}{2}, \frac{3}{4}\right)$ [from $x - 2y = 2$ and $3x + 2y = 12$ together: adding, $4x = 14$], $\left(\frac{3}{2}, \frac{15}{4}\right)$ [from $3x + 2y = 12$ and $-3x + 2y = 3$ together: subtracting, $6x = 9$], and $(0, 1.5)$ [from $-3x + 2y = 3, x = 0$] – each checked against the remaining constraints and confirmed feasible.

Evaluating $Z = 5x + 2y$: $(0, 0) \to 0$; $(2, 0) \to 10$; $\left(\frac{7}{2}, \frac{3}{4}\right) \to 17.5 + 1.5 = 19$; $\left(\frac{3}{2}, \frac{15}{4}\right) \to 7.5 + 7.5 = 15$; $(0, 1.5) \to 3$.

Final Answer

Z is maximum at $\left(\frac{7}{2}, \frac{3}{4}\right)$ and the maximum value of Z is 19.

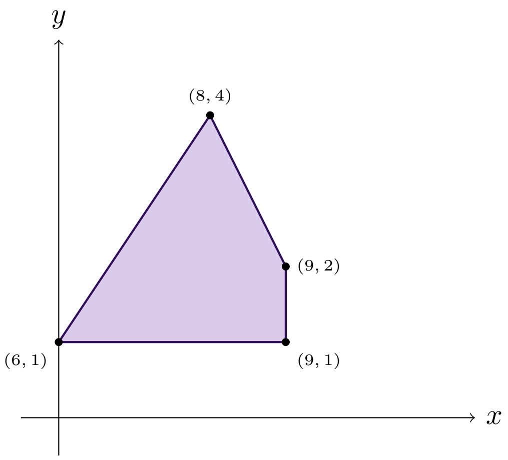
Quick Recall: With 3 or more $\leq$ constraints, the feasible region is a polygon with that many possible sides – systematically pair every two boundary lines, discard infeasible intersections, then evaluate Z only at survivors.

Type 10

A maximize/minimize LPP where, in addition to the usual constraint lines, the variables themselves are directly bounded by extra lines like $x \leq k$ or $y \geq k$ – these act as simple vertical/horizontal boundary lines that can cut the polygon differently from a normal constraint, and are easy to miss since they look like an afterthought at the end of the constraint list.

Source: Exercise 12.2, Q7(ii)

Given: Maximise $Z = 4x - 3y$ subject to $2x + y \leq 20$, $3x - 2y \geq 16$, $x \leq 9$, $y \geq 1$.



Feasible region for Type 10 – a quadrilateral cut off by the direct bounds $x \leq 9$ and $y \geq 1$; maximum occurs at $(9, 1)$.

Solution:

At $y = 1$ (the lower bound line): $3x - 2(1) = 16 \Rightarrow x = 6$; and $2x + 1 \leq 20 \Rightarrow x \leq 9.5$; combined with the direct bound $x \leq 9$, the tighter cap is $x = 9$. So on this edge, the corners are $(6, 1)$ [from $3x - 2y = 16$] and $(9, 1)$ [from the direct bound $x = 9$, which is tighter than the 9.5 given by $2x + y = 20$].

At $x = 9$ (the direct bound): $y \geq 1$, and $2(9) + y \leq 20 \Rightarrow y \leq 2$, giving the corner $(9, 2)$.

$2x + y = 20$ and $3x - 2y = 16$ together: multiplying the first by 2 and adding, $7x = 56 \Rightarrow x = 8,\ y = 4$ – checking $x \leq 9$ and $y \geq 1$ both hold, giving $(8, 4)$.

Corner points, in order around the region: $(6, 1), (9, 1), (9, 2), (8, 4)$. Evaluating $Z = 4x - 3y$:

$$Z(6, 1) = 21, \quad Z(9, 1) = 33, \quad Z(9, 2) = 30, \quad Z(8, 4) = 20$$

The largest is 33.

Final Answer

Z is maximum at the point $(9, 1)$ and the maximum value of Z is 33.

Common Mistake: Ignoring the direct bound $x \leq 9$ once the "main" constraint $2x + y \leq 20$ has already been used, and instead running the boundary edge all the way to $x = 9.5$ – whichever bound is TIGHTER (smaller for an upper bound) is the one that actually cuts the feasible region.

Quick Recall: Direct bounds like $x \leq k$ or $y \geq k$ are extra straight boundary lines just like any constraint – draw them in and always use whichever bound is tighter at that edge.

III. Special Cases in Graphical Method

Type 11

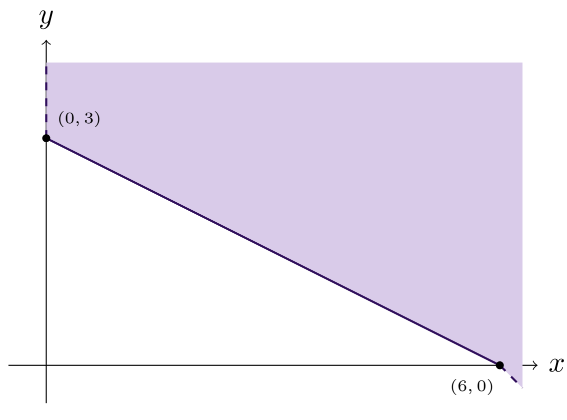
A minimize or maximize LPP where TWO different corner points give the exact same optimal value of Z – this happens precisely when the objective function's line is PARALLEL to one edge of the feasible region. When this occurs, the optimal value is not achieved at just those two points but at EVERY point on the line segment joining them, giving infinitely many optimal solutions.

★ **Board Favorite** – Boards frequently ask you to explicitly "show that the minimum/maximum occurs at more than two points" – recognising the parallel-line condition (rather than just reporting two equal numbers) is exactly what earns the reasoning marks here.

Formula Used: $Z(P_1) = Z(P_2) \Rightarrow Z$ is optimal at every point of segment P_1P_2

Source: Exercise 12.2, Q5

Given: Minimize $Z = x + 2y$ subject to $2x + y \geq 3$, $x + 2y \geq 6$, $x \geq 0$, $y \geq 0$. Show that the minimum value of Z occurs at more than two points.



Feasible region for Type 11 – unbounded, with minimum achieved along the entire edge from $(6, 0)$ *to* $(0, 3)$.

Solution:

Find the corner points: $x + 2y = 6$ with $y = 0$ gives $x = 6$ – checking $2x + y = 12 \geq 3$ holds, so $(6, 0)$.

$2x + y = 3$ with $x = 0$ gives $y = 3$; checking $x + 2y = 6 \geq 6$ holds (equality), so $(0, 3)$.

$2x + y = 3$ and $x + 2y = 6$ together: from the first, $y = 3 - 2x$; substituting, $x + 2(3 - 2x) =$

$6 \Rightarrow -3x = 0 \Rightarrow x = 0, y = 3$ – the two lines meet exactly at $(0, 3)$ itself, so there is no third, separate intersection corner between them.

So the only two corner points of this unbounded region are $(6, 0)$ and $(0, 3)$. Evaluating $Z = x + 2y$:

$$Z(6, 0) = 6, \qquad Z(0, 3) = 0 + 6 = 6$$

Both corner points give the SAME value, $Z = 6$. Checking whether the objective line is parallel to the edge joining them: the edge from $(6, 0)$ to $(0, 3)$ lies exactly on the line $x + 2y = 6$ – and this is the SAME expression as the objective function $Z = x + 2y$ itself, so $Z = 6$ for every single point on that segment, not just the two ends.

Final Answer

Minimum $Z = 6$ occurs at $(6, 0)$ and $(0, 3)$; in fact, every point on the line segment joining these two points gives the same minimum value 6.

Common Mistake: Stopping after finding $Z(6, 0) = Z(0, 3) = 6$ and just calling that "the minimum" without the extra sentence explaining WHY – the reasoning mark specifically wants the observation that the edge $x + 2y = 6$ is identical in form to the objective function, hence infinitely many solutions.

Quick Recall: Whenever two corner points give equal Z-values, check if the edge joining them lies on a line parallel to (or the same as) the objective function – if so, the whole edge is optimal, not just its endpoints.

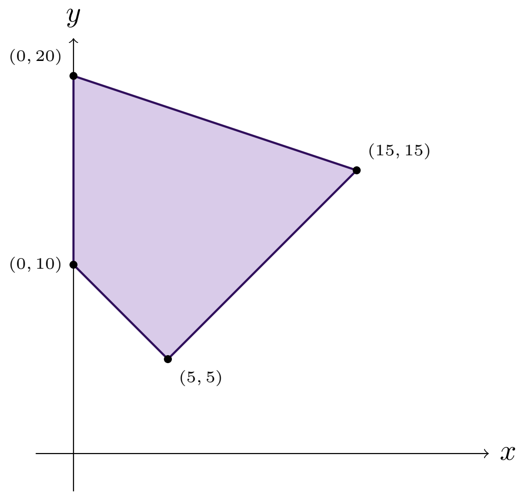
Type 12

A single feasible region where the question asks for BOTH the minimum AND the maximum of the same objective function Z over the same constraints – you build one feasible region, then simply read off the smallest and largest Z-values among all its corner points in one pass.

★ Board Favorite – This "minimize and maximize together" phrasing is an NCERT-and-ISC staple – it tests whether a student can evaluate every corner correctly rather than stopping the moment one extreme is found, and it commonly comes bundled with the multiple-optimal-solution trap from Type 11 at the maximum end.

Source: Exercise 12.2, Q6 (NCERT)

Given: Minimize and maximize $Z = 3x + 9y$ subject to $x + y \geq 10$, $x + 3y \leq 60$, $x \leq y$, $x \geq 0$, $y \geq 0$.



Feasible region for Type 12 – quadrilateral $(0, 10), (5, 5), (15, 15), (0, 20)$*; minimum at* $(5, 5)$*, maximum along the edge from* $(15, 15)$ *to* $(0, 20)$*.*

Solution:

Find corner points: $x + y = 10$ and $x = y$ together give $x = y = 5$, so $(5, 5)$; checking $x + 3y = 20 \leq 60$ confirms feasible.

$x + 3y = 60$ and $x = y$ together give $x = y = 15$, so $(15, 15)$; checking $x + y = 30 \geq 10$ confirms feasible.

$x + y = 10$ with $x = 0$: gives $y = 10$, so $(0, 10)$; checking $x \leq y$ $(0 \leq 10)$ and $x + 3y = 30 \leq 60$ both hold.

$x + 3y = 60$ with $x = 0$: gives $y = 20$, so $(0, 20)$; checking $x \leq y$ and $x + y = 20 \geq 10$ both hold.

Corner points: $(0, 10), (5, 5), (15, 15), (0, 20)$. Evaluating $Z = 3x + 9y$:

$$Z(0, 10) = 90, \quad Z(5, 5) = 60, \quad Z(15, 15) = 180, \quad Z(0, 20) = 180$$

The smallest is 60 (a unique minimum). The largest is 180, achieved at BOTH $(15, 15)$ and $(0, 20)$ – checking, the edge between them lies on $x+3y = 60$, and since $Z = 3x+9y = 3(x + 3y)$, this equals $3(60) = 180$ everywhere on that edge, confirming infinitely many maximizing points too.

Final Answer

Minimum $Z = 60$ at $(5, 5)$. Maximum $Z = 180$ occurs at both $(15, 15)$ and $(0, 20)$, and in fact at every point on the segment joining them.

Common Mistake: Reporting only the minimum (or only the maximum) because the question is skimmed quickly – this type always demands BOTH extremes from the same table of corner-point values, so evaluate the full table once and read off both ends.

Quick Recall: Build the feasible region once, evaluate Z at every corner in a single table, then read the smallest value as the minimum and the largest as the maximum – no need to solve the LPP twice.

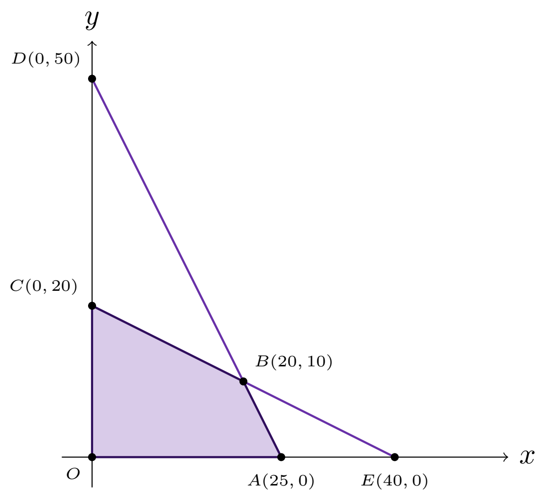
Type 13

The feasible region is already drawn for you as a graph with labelled points, and you are asked to work BACKWARDS: read off the two boundary lines from the labelled intercepts, write them as inequality constraints, identify a labelled intersection point algebraically, and then evaluate the objective function at the ACTUAL feasible region's corners (which may be different from the labelled reference points on the graph).

★ **Board Favorite –** This "read the graph, don't draw it" format is exactly how recent ISC Specimen Papers have started testing LPP – it rewards students who can extract equations of lines from two given points quickly, which is a skill often neglected when everyone practices only the "solve from scratch" types.

Source: Exercise 12.2, Q24 (ISC Specimen Paper 2026)

Given: The feasible region for an LPP is shown with reference points $D(0, 50)$, $C(0, 20)$, $A(25, 0)$, $E(40, 0)$, and an unlabelled intersection point B. (a) Write the constraints. (b) Find the coordinates of B. (c) Find the maximum value of $Z = x + y$.



Reading a given feasible region – shaded quadrilateral $O(0, 0), A(25, 0), B(20, 10), C(0, 20)$, with reference lines through $D(0, 50)$–$A(25, 0)$ and $C(0, 20)$–$E(40, 0)$.

Solution:

(a) Finding the constraint lines: the line through $D(0, 50)$ and $A(25, 0)$ has slope $\dfrac{0-50}{25-0} = -2$, giving the equation $y = 50 - 2x$, i.e. $2x + y = 50$; since the shaded region lies below/inside this line, the constraint is $2x + y \le 50$.

The line through $C(0, 20)$ and $E(40, 0)$ has slope $\dfrac{0-20}{40-0} = -\frac{1}{2}$, giving $y = 20 - \frac{1}{2}x$, i.e. $x + 2y = 40$; the constraint is $x + 2y \le 40$.

Together with non-negativity, the constraints are $2x + y \le 50$, $x + 2y \le 40$, $x \ge 0$, $y \ge 0$.

(b) Finding B: B is the intersection of the two boundary lines. Multiplying $2x+y=50$ by 2 gives $4x+2y=100$; subtracting $x+2y=40$ gives $3x=60 \Rightarrow x=20$, and then $y=50-40=10$.

(c) Maximizing $Z=x+y$: the TRUE feasible region's corners are $O(0,0)$, $A(25,0)$ (the x-intercept of the binding constraint $2x+y=50$, since it gives a smaller x than E's 40), $B(20,10)$, and $C(0,20)$ (the y-intercept of the binding constraint $x+2y=40$, smaller than D's 50). Evaluating:

$$Z(0,0)=0, \quad Z(25,0)=25, \quad Z(20,10)=30, \quad Z(0,20)=20$$

Final Answer

(a) $2x+y \leq 50,\ x+2y \leq 40,\ x \geq 0,\ y \geq 0.$ (b) $B=(20,10)$. (c) Maximum value of $Z=30$.

Common Mistake: Using the labelled reference points $D(0,50)$ and $E(40,0)$ as if they were corners of the FEASIBLE region – they are just the axis-intercepts of the individual lines used to plot the graph; the actual shaded region's corners are A and C (whichever intercept is smaller/more restrictive on each axis), plus O and B.

Quick Recall: To read constraints off a graph: find the equation of each boundary line from its two labelled intercepts using slope-intercept form, then decide $\leq$ or $\geq$ from which side is shaded.

Type 14

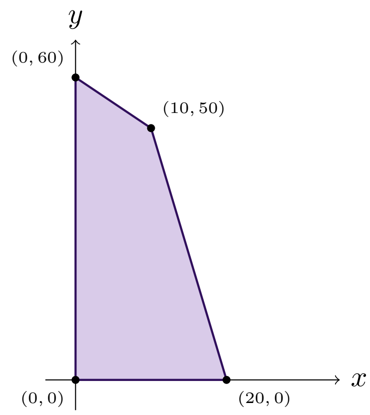
The objective function itself contains unknown positive parameters ($Z=px+qy$) instead of fixed numbers, and you're told the maximum (or minimum) occurs at two SPECIFIC given corner points. You must set the two corresponding Z-expressions equal to each other (since both give the same optimal value) and solve for the relationship between p and q – this is the algebraic mirror of the multiple-optimal-solutions idea from Type 11, but run in reverse.

★ **Board Favorite –** This "find the relation between p and q" question format has appeared in ISC Specimen Papers as a stand-alone twist on the standard LPP – it tests conceptual understanding of WHY multiple optimal solutions occur, not just mechanical corner-point evaluation.

Formula Used: $Z(P_1)=Z(P_2) \Rightarrow$ equate the two expressions in p,q and simplify

Source: Exercise 12.2, Q26 (ISC Specimen Paper 2025)

Given: $Z=px+qy$ where $p,q>0$, subject to $x+y \leq 60$, $5x+y \leq 100$, $x \geq 0$, $y \geq 0$. (i) Solve graphically to find the corner points. (ii) If Z is maximum at $(0,60)$ and $(10,50)$, find the relation between p and q, and the number of optimal solutions.



Feasible region for Type 14 – quadrilateral $(0,0),(20,0),(10,50),(0,60)$*;*
Z *is maximum along the entire edge from* $(10,50)$ *to* $(0,60)$*.*

Solution:

(i) Corner points: $5x+y=100$ with $y=0$ gives $x=20$; checking $x+y=20\le 60$ holds, so $(20,0)$ – note $x+y=60, y=0$ would give $(60,0)$, but that fails $5x+y=300\le 100$, so it isn't a true vertex.

$x+y=60$ and $5x+y=100$ together: subtracting, $4x=40 \Rightarrow x=10,\ y=50$, giving $(10,50)$.

$x+y=60$ with $x=0$ gives $y=60$; checking $5(0)+60=60\le 100$ holds, so $(0,60)$.

Corner points: $(0,0),(20,0),(10,50),(0,60)$.

(ii) Relation between p,q**:** since $Z=px+qy$ is maximum at BOTH $(0,60)$ and $(10,50)$, these must give equal Z-values:

$$Z(0,60)=60q, \qquad Z(10,50)=10p+50q$$

Setting them equal:

$$60q=10p+50q \ \Rightarrow\ 10q=10p \ \Rightarrow\ p=q$$

Since the maximum occurs at two distinct corner points with $p=q$, by the multiple-optimal-solutions principle it occurs at every point on the segment joining $(0,60)$ and $(10,50)$.

Final Answer

(i) Corner points: $(0,0),(20,0),(10,50),(0,60)$. (ii) $p=q$; the LPP has an infinite number of optimal solutions.

Common Mistake: Trying to find numerical values of p and q individually – the question only ever asks for the RELATION between them (like $p=q$), since with two unknowns and one equation, individual values can never be pinned down.

Quick Recall: "Maximum at two given points" with unknown coefficients $\Rightarrow$ set the two Z-expressions equal and simplify to find the relation between the coefficients – this always signals infinitely many optimal solutions.

IV. Applied Word Problems – Formulate & Solve

Type 15

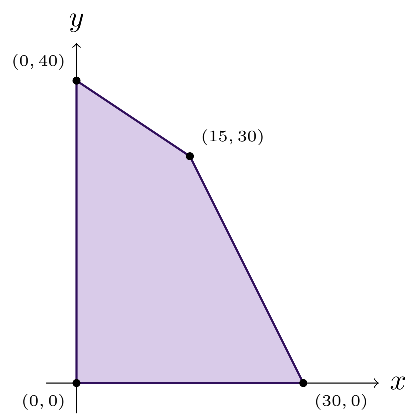
The full end-to-end LPP word problem: two products, each consuming two different resources (machine time, raw material, budget) at different fixed rates, each earning a fixed profit, with resource caps given. You must BOTH formulate the LPP (as in Section I) AND solve it graphically (as in Section II) in one continuous answer – this is the single most common full-mark LPP question type on ISC papers, and Ex 12.2 Q9–18 are all variations of this exact pattern.

★ **Board Favorite** – This is the highest-frequency LPP pattern on ISC boards – almost every year's paper has one "two products, two resources, maximize profit" question, and Ex 12.2 alone repeats it across ten questions (Q9–18) with different product/resource stories.

Formula Used: Profit per unit $\times$ (units of product 1) $+$ Profit per unit $\times$ (units of product 2) $= Z$ (maximize)

Source: Exercise 12.2, Q13 (ISC 2022)

Given: Machines X and Y run at most 360 minutes/day. Toy A needs 12 min on X and 6 min on Y; Toy B needs 6 min on X and 9 min on Y. Profits: Rs. 30 on Toy A, Rs. 20 on Toy B.



Feasible region for Type 15 – quadrilateral $(0,0), (30,0), (15,30), (0,40)$*; maximum profit at* $(15,30)$*.*

Solution:

Let x = number of Toy A and y = number of Toy B manufactured per day.

Machine X constraint: $12x + 6y \leq 360 \Rightarrow 2x + y \leq 60$.

Machine Y constraint: $6x + 9y \leq 360 \Rightarrow 2x + 3y \leq 120$.

Objective: $Z = 30x + 20y$ (maximize).

Corner points: $2x + y = 60$ with $y = 0$ gives $x = 30$; checking $2x + 3y = 60 \leq 120$ holds, so $(30, 0)$.

$2x + y = 60$ and $2x + 3y = 120$ together: subtracting, $2y = 60 \Rightarrow y = 30, x = 15$, giving $(15, 30)$.

$2x + 3y = 120$ with $x = 0$ gives $y = 40$; checking $2(0) + 40 = 40 \leq 60$ holds, so $(0, 40)$.

Evaluating $Z = 30x + 20y$ at $(0, 0), (30, 0), (15, 30), (0, 40)$: $0,\ 900,\ 1050,\ 800$.

Final Answer

Maximum profit of Rs. 1050 occurs when 15 toys of Type A and 30 toys of Type B are manufactured.

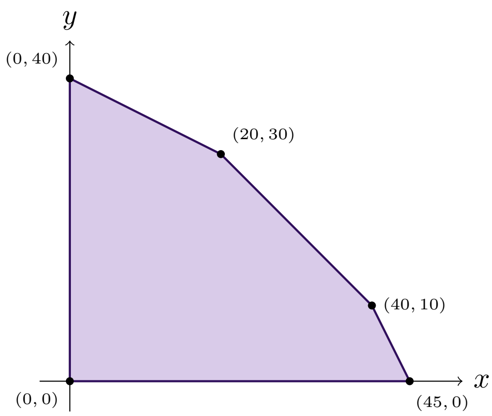
Common Mistake: Forgetting that this type needs the FULL two-part answer – both the constraint formulation AND the corner-point evaluation – boards award marks separately for each stage, so writing only the final numbers without showing the formulation loses marks even if the answer is correct.

Note: Ex 12.2 Q9–12 and Q14–17 all follow this exact same "two products, two resource constraints, maximize profit" skeleton with different stories (workers/capital, nuts/bolts, rings/chains, gold/silver, sewing machines, souvenirs, soaps, lamps/shades) – once this method is solid, every one of those solves the same way.

Alternate Board-Style Example

Source: Exercise 12.2, Q18 (ISC 2019)

Given: A carpenter has 90, 80, 50 running feet of teak wood, plywood, rosewood. Product A needs 2, 1, 1 running feet respectively; Product B needs 1, 2, 1 running feet respectively. A sells for Rs. 48/unit, B for Rs. 40/unit.



Feasible region for the ISC 2019 alternate example – pentagon $(0, 0), (45, 0), (40, 10), (20, 30), (0, 40)$*; maximum revenue at* $(40, 10)$.

Solution:

Let $x, y =$ units of Products A, B. Constraints: teak $2x + y \le 90$, plywood $x + 2y \le 80$, rosewood $x + y \le 50$, $x, y \ge 0$. Objective: $Z = 48x + 40y$.

Testing pairwise intersections against all three constraints gives corner points $(0,0)$, $(45,0)$, $(40,10)$, $(20,30)$, $(0,40)$ – note the intersection of "teak" and "plywood" alone, $(33.3, 23.3)$, fails the rosewood constraint $(x + y = 56.7 > 50)$ and is correctly excluded.

Evaluating $Z = 48x + 40y$: $(0,0) \to 0$; $(45,0) \to 2160$; $(40,10) \to 2320$; $(20,30) \to 2160$; $(0,40) \to 1600$.

Final Answer

40 units of Product A and 10 units of Product B; maximum revenue $= Rs.\,2320$.

Quick Recall: Two products × two-or-more resources, maximize profit: write one $\le$ constraint per resource (using per-unit consumption), objective = sum of (profit per unit × quantity), then run the standard corner-point method.

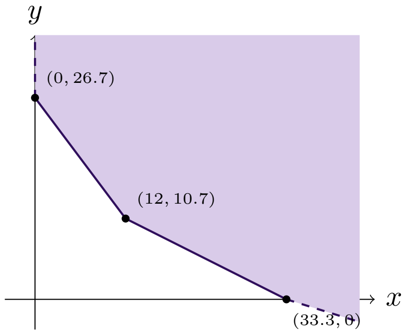
Type 16

The mixture/diet-type LPP: two "ingredients," each contributing a fixed amount toward two or more nutritional/quality requirements, each costing a fixed price, and you must meet AT LEAST the minimum requirement of each nutrient while minimizing total cost. The constraints are $\ge$ (not $\le$), giving an unbounded-region minimize problem, structurally identical to Type 8 but wrapped in a full word-problem formulation.

★ **Board Favorite** – Diet/mixture-minimization word problems are the "mirror image" of the Type 15 profit-maximization word problems and appear just as often – Ex 12.2 Q19–22 all repeat this pattern with different nutrient/ingredient stories, including a genuine ISC 2017 board question.

Source: Exercise 12.2, Q19

Given: At least 80 units of Vitamin A and 100 units of minerals needed. Food F_1 (Rs. 5/unit) has 4 units Vit A, 3 units minerals. Food F_2 (Rs. 6/unit) has 3 units Vit A, 6 units minerals.



Feasible region for Type 16 – unbounded, above the line joining $(0, \frac{80}{3})$, $(12, \frac{32}{3})$, $(\frac{100}{3}, 0)$; minimum cost at $(12, \frac{32}{3})$.

Solution:

Let x, y = units of F_1, F_2 used. Constraints: $4x + 3y \geq 80$ (Vit A), $3x + 6y \geq 100$ (minerals), $x, y \geq 0$. Objective: $Z = 5x + 6y$ (minimize).

Corner points: $4x + 3y = 80$ with $x = 0$ gives $y = \frac{80}{3}$; checking $3(0) + 6(\frac{80}{3}) = 160 \geq 100$ holds, so $\left(0, \frac{80}{3}\right)$.

$4x + 3y = 80$ and $3x + 6y = 100$ together: doubling the first, $8x + 6y = 160$; subtracting the second, $5x = 60 \Rightarrow x = 12,\ y = \frac{32}{3}$, giving $\left(12, \frac{32}{3}\right)$.

$3x + 6y = 100$ with $y = 0$ gives $x = \frac{100}{3}$; checking $4(\frac{100}{3}) = 133.3 \geq 80$ holds, so $\left(\frac{100}{3}, 0\right)$.

Evaluating $Z = 5x + 6y$: $\left(0, \frac{80}{3}\right) \to 160$; $\left(12, \frac{32}{3}\right) \to 60 + 64 = 124$; $\left(\frac{100}{3}, 0\right) \to 166.7$.

Final Answer

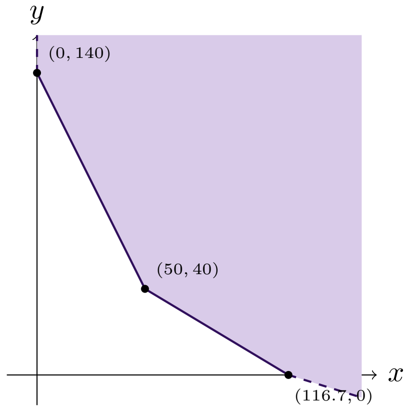
Minimum cost of the diet is Rs. 124 when 12 units of Food F_1 and $\frac{32}{3}$ units of Food F_2 are mixed.

Common Mistake: On a $\geq$-constrained (unbounded) minimize problem, evaluating Z only at the axis intercepts and skipping the interior intersection point – the true minimum on this type of region is almost always at the "inner corner," not at an axis intercept.

Alternate Board-Style Example

Source: Exercise 12.2, Q22 (ISC 2017)

Given: Fertiliser type A (10% nitrogen, 6% phosphoric acid, Rs. 5/kg) and type B (5% nitrogen, 10% phosphoric acid, Rs. 8/kg). At least 7 kg nitrogen and 7 kg phosphoric acid required.



Feasible region for the ISC 2017 alternate example – unbounded, minimum cost at $(50, 40)$*.*

Solution:

Let $x, y =$ kg of type A, B used. Nitrogen: $0.10x + 0.05y \geq 7 \Rightarrow 2x + y \geq 140$. Phosphoric acid: $0.06x + 0.10y \geq 7 \Rightarrow 3x + 5y \geq 350$. Objective: $Z = 5x + 8y$ (minimize).

Corner points: $(0, 140)$ [from $2x + y = 140, x = 0$]; $\left(\frac{350}{3}, 0\right)$ [from $3x + 5y = 350, y = 0$]; and $2x + y = 140$ with $3x + 5y = 350$ together: from the first $y = 140 - 2x$, substituting gives $3x + 5(140 - 2x) = 350 \Rightarrow -7x = -350 \Rightarrow x = 50,\ y = 40$, giving $(50, 40)$.

Evaluating $Z = 5x + 8y$: $(0, 140) \to 1120$; $(50, 40) \to 250 + 320 = 570$; $\left(\frac{350}{3}, 0\right) \to 583.3$.

Final Answer

50 kg of type A and 40 kg of type B; minimum cost $= Rs.\,570$.

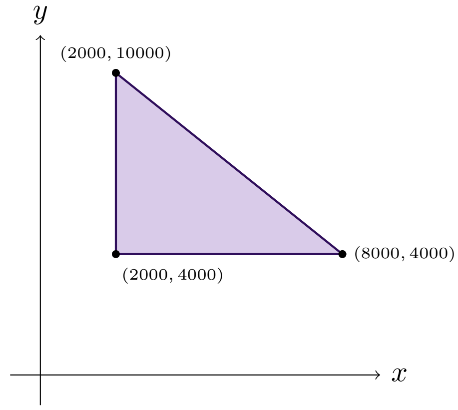
Quick Recall: Diet/mixture minimize problems: convert every percentage requirement into a $\geq$ constraint, objective = sum of (cost per unit $\times$ quantity), and the minimum on this unbounded region is almost always at the interior intersection corner.

Type 17

An investment-allocation LPP with a total budget cap ($\leq$) combined with individual minimum-investment requirements per option ($\geq$) – this mixed constraint structure (one $\leq$, two $\geq$) produces a small triangular feasible region, and the objective is to MAXIMIZE return (unlike Type 16's minimize-cost mixed-constraint problems).

Source: Exercise 12.2, Q23

Given: David invests at most Rs. 12000 in Bonds A and B. At least Rs. 2000 in Bond A and at least Rs. 4000 in Bond B. Interest rates: 8% on A, 10% on B.



Feasible region for Type 17 – triangle $(2000, 4000), (2000, 10000), (8000, 4000)$*; maximum interest at* $(2000, 10000)$.

Solution:

Let $x, y =$ amounts (Rs.) invested in Bonds A, B. Constraints: $x + y \leq 12000$, $x \geq 2000$, $y \geq 4000$. Objective: $Z = 0.08x + 0.10y$ (maximize).

Corner points: $x = 2000, y = 4000$ together give the point $(2000, 4000)$ directly, since both bounds are simultaneously tight.

$x = 2000$ with $x + y = 12000$ gives $y = 10000$, so $(2000, 10000)$.

$y = 4000$ with $x + y = 12000$ gives $x = 8000$, so $(8000, 4000)$.

Evaluating $Z = 0.08x + 0.10y$: $(2000, 4000) \to 160 + 400 = 560$; $(2000, 10000) \to 160 + 1000 = 1160$; $(8000, 4000) \to 640 + 400 = 1040$.

Final Answer

Rs. 2000 in Bond A and Rs. 10000 in Bond B yield the maximum interest of Rs. 1160.

Common Mistake: Confusing this with a Type 16-style minimize problem just because it has $\geq$ constraints – always check the objective's direction (maximize interest here) rather than assuming from the constraint types alone.

Quick Recall: Total-budget cap ($\leq$) plus per-option minimums ($\geq$) gives a small triangle whose three corners come directly from pairing the bounds two at a time – evaluate Z at all three and pick the extreme the question actually asks for.